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Monitoring of PAHs and alkylated PAHs in aquatic organisms after 1 month from the *Solar I* oil spill off the coast of Guimaras Island, Philippines

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Abstract Following the oil spill accident of the *Solar I* tanker in 2006 off the coast of Guimaras Island in the Philippines, polycyclic aromatic hydrocarbons (PAHs) and alkylated PAHs in some aquatic organisms were investigated at Luzaran in Guimaras and Taklong Islands, which were heavily polluted with spilled oil, immediately and 1 month after the accident. The concentrations of total PAHs were 11.9–52.3 ng/g dry weight in fish. Meanwhile, total PAH concentrations in shellfish were 38.0–3,102 ng/g dry weight in Luzaran and 128–236 ng/g dry weight in Taklong. Pyrene, phenanthrene, and fluoranthene were dominant in most fish and chrysene in all shellfish. Significantly higher concentrations of all alkylated homologs were detected in shellfish than in fish. These differences had two possible causes, that is, the differences between fish and shellfish could be attributed to the uptake routes and/or their metabolizing abilities.

Keywords Oil spill · PAHs · Alkylated PAHs · Aquatic organisms · *Solar I* · Guimaras Island

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Introduction

On August 11, 2006, the *Solar I* tanker was hit by hard rainstorm and sank off the coast of Guimaras Island in the Philippines. About 350,000 tons of heavy fuel oil was spilled from the tanker polluting about 200 km of shoreline from Guimaras to Panay Island and Negros Island. A large extent of shoreline in Guimaras was especially polluted with heavy oil, and coastal fisheries were seriously damaged. Over 450 ha of huge protected

mangroves with complex topography at Guimaras was also contaminated. The large mass of oil was manually removed over a 1-month period, but several sites of polluted shoreline were left in an "unclean" situation.

In general, spilled oil degraded with several factors such as evaporation, diffusion, photo-oxidation, and biodegradation (Hayes and Michel 1999; Laffon et al. 2006; Michel and Hayes 1999; Owens et al. 2008; Taylor and Reimer 2008). In tropical areas, because of strong sunshine, high temperature, and frequent storms, spilled oil is expected to degrade faster than in the temperate and boreal areas although it still is poorly studied.

Since polycyclic aromatic hydrocarbons (PAHs) are persistent in the environment (Michel and Hayes 1999; Wang et al. 2001; Wetzel and van Vleet 2003), their toxicities to aquatic organisms have been examined (e.g., Barron et al. 2004; Francioni et al. 2007; Incardona et al. 2004; Reynaud and Deschaux 2006). The acute toxicity of PAHs to fish generally induces narcosis, which is a reversible anesthetic effect (Barron et al. 2004). In addition, acute PAH exposures to early life stages of fish resulted in developmental abnormalities (Pollino and Holdway 2002). Alkylated PAHs are also included in the oil as well as parent PAHs, and their toxicities to the early life stage of fish, mainly embryo development, are also suspected (Heintz et al. 2000; Rhodes et al. 2005). When an oil spill occurs in a water system, the concentrations of PAHs and alkylated PAHs are significantly high in water and could have significant threats and several negative impacts on the surrounding aquatic organisms.

In the present study, since the initial impacts of the *Solar I* oil were investigated, the accumulations of PAHs and alkylated PAHs were examined in some species of fish, shellfish, and crabs immediately after the spill and 1 month later.

Materials and methods

Collection of organisms

The first investigation was conducted around the southern tip of Guimaras Island, Luzaran (Fig. 1), 1 week after the spill (August 2006), when some species of fish were collected (*Tropidnius* sp., *Lethrinus harak*, *Evynnis* sp., *Caranx* sp., and an unknown species 1). Each of the fish species was collected except for *Evynnis* sp. About 1 month after the accident (September 2006), the second investigation was conducted at Luzaran and the north-east part of Taklong Island, next to Guimaras Island (Fig. 1). At that time, the following species were collected: shellfish—*Trichomya* sp., *Monilea* sp., *Cypraea* sp., *Clypeomorus* sp., *Gafrarium* sp., *Pinctada* sp., *Siratus* sp.; oyster—(unknown species), *Nerita* sp.; crab—*Scylla* sp.; fish—unknown species 2. After the collections, all samples were stored at −20°C until analysis.

Extractions of PAHs and alkylated PAHs

The shellfish samples were thawed and shucked. One sample for analysis was composed of about 5 g wet weight of combined soft tissues of several individuals for each species. Two samples were prepared for one species of shellfish, although some samples were prepared from only one sample for analysis because sometimes only a small number of individuals on one species were collected from a sampling site. Individual whole body fish without the heads and bones were singly used for analysis. Crabs were thawed, shucked, and analyzed individually. All biological samples were freeze-dried. Freeze-dried subsamples (equivalent to 3 g of wet weight) of shellfish were ultrasonically extracted with 20 mL of dichloromethane–hexane mixture (1:1, *v*/*v*) for 15 min. After the extraction, centrifuging at 760×*g* for 10 min, the supernatant was collected. These procedures were repeated twice and the extracts were combined. The combined extract was concentrated to 0.2 mL under a nitrogen stream. The concentrated extract was added to 6 mL of 1 M potassium hydrate–ethanol and saponified in successive circulating solution under 90°C for 1 h. The solution was added to 20 mL of distilled water and 10 mL of dichloromethane–hexane mixture (1:3, *v*/*v*). After shaking and centrifuging at 760×*g* for 10 min, the upper phase was collected. The procedure was repeated twice, and the collected solutions were combined. The combined solution was completely exchanged with a

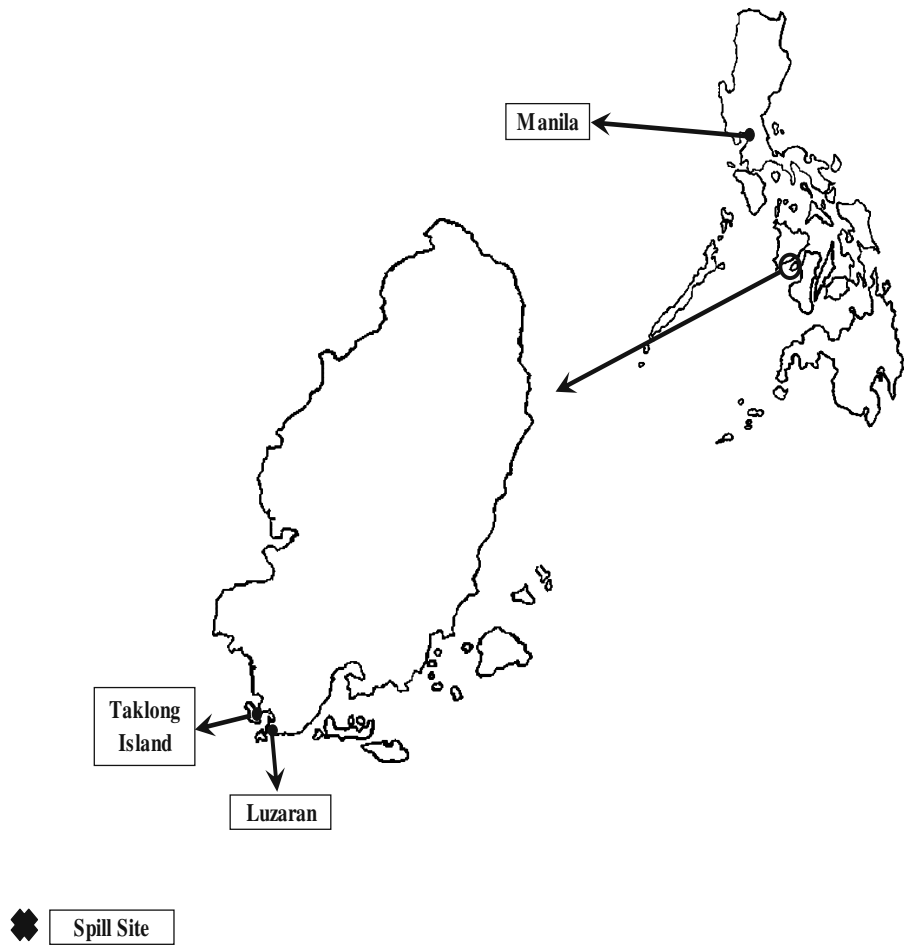


Fig. 1 Locations of the sampling site where aquatic organisms were collected

hexane solvent. The hexane solution was concentrated to 1 mL. The concentrated hexane solution was loaded on a silica-gel column (packed in a Pasteur pipette), washed with 10 mL hexane, and eluted with 10 mL hexane and 10 mL 1% acetone–hexane. Their eluents were combined, spiked with internal standards (naphthalene-d_8, anthracene-d_{10}, pyrene-d_{10}, and chrysene-d_{12}), and concentrated until 0.2 mL with nitrogen stream.

Gas chromatographic analysis

Measurements of PAHs and alkylated PAHs were conducted, using a gas chromatograph (Agilent 6890; Agilent Technologies, USA) equipped with a mass spectrometer (Agilent 5973 MSD). One microliter of the sample was injected by an Agilent 7683B automatic sampler into a DB-5MS (0.25 mm i.d. × 60 m, film thickness 0.25 μm; J&W Scientific, USA). The injector and detector temperatures were 270°C. The oven temperature of the column was incrementally increased as follows: 60°C for 1 min, 60°C to 200°C at a rate of 20°C/min, maintained at 200°C for 1 min, 200°C to 290°C at a rate of 3°C/min, and then maintained at 290°C for 5 min. The quantities of the target PAHs were performed in SIM mode using standard solution of 18 PAHs standard mixture (naphthalene, acenaphthylene, acenaphthene, fluorene, dibenzothiophene, phenanthrene, anthracene, fluoranthene, benz(*a*)anthracene, chrysene, benzo(*b*)fluoranthene, benzo(*k*)fluoranthene, benzo(*a*)pyrene, perylene, dibenz(*a*,*h*)-anthracene, indeno(1,2,3-*cd*)pyrene, and benzo(*g*,*h*,*i*)perylene. The index total PAHs were calculated as the sum of the total quantity of all these compounds. Alkylated PAHs of naphthalene (C1 to C4), fluorene (C1 to C3),

phenanthrene/anthracene (C1 to C4), pyrene/fluoranthene (C1 to C3), and chrysene/benzo(*a*)anthracene (C1 to C3) were quantitated using straight baseline integration of each level alkylation (Wang et al. 2007). All PAHs and alkylated PAHs in the individual sample were simultaneously analyzed in a single GC/MS measurement.

Results and discussions

In September 2006, 1 month after the oil spill, a great amount of spilled oil was left extensively in Guimaras beach. The oil might have been left in sea water and sediments at some place during several months that follow. As a result, aquatic organisms made their habitat around those "hot spots", accumulating heavy oil components.

PAH

The concentrations and distribution patterns of PAHs were significantly different among fish, shellfish, and crabs. The concentrations of total PAHs in fish, collected immediately and 1 month after the oil spill, were 11.9–52.3 ng/g dry weight, and the concentration levels in fish were similar even though 1 month had passed (Fig. 2). Meanwhile, total PAH concentrations in shellfish were much higher than those in fish. They were 38.0–3,102 ng/g dry weight in Luzaran (Fig. 3) and 128–236 ng/g dry weight in Taklong (Fig. 4). Concentrations in *Monilea* sp., *Clypeomorus* sp., and *Pinctada* sp. at Luzaran were significantly higher at 3,075, 1,227, and 3,102 ng/g dry weight, respectively (Fig. 3), although 1 month had passed after the *Solar I* oil spill. Before the *Solar I* oil spill, we investigated some contaminants including PAHs in bivalves (brown mussel (*Perna* sp.), oyster (unknown species), and green mussel (*Perna viridis*)) in Panay Island, which is next to Guimaras (Uno et al. 2007), and total PAHs in these bivalves were from 2.9 to 4.2 ng/g wet weight (they were roughly estimated under 20 ng/g dry weight). Guimaras Island has a smaller population, less traffic, and industries, and it was highly possible that PAH concentrations at Guimaras before the spill were similar with or less than those in Panay. In the present study, PAH concentrations in shellfish at Luzaran and Taklong were higher than those in Panay before the spill, and those PAHs could be derived from the *Solar I* oil spill. Nieto et al. (2006) reported that the maximum concentration of total PAH (13 PAHs) was 5.9 mg/kg dry weight in *Mytilus galloprovincialis* 2 months after the *Prestige* oil spill. Soriano et al. (2006) also reported that the concentration was above 1,000 to 7,780 μg/kg dry weight in *M. galloprovincialis*

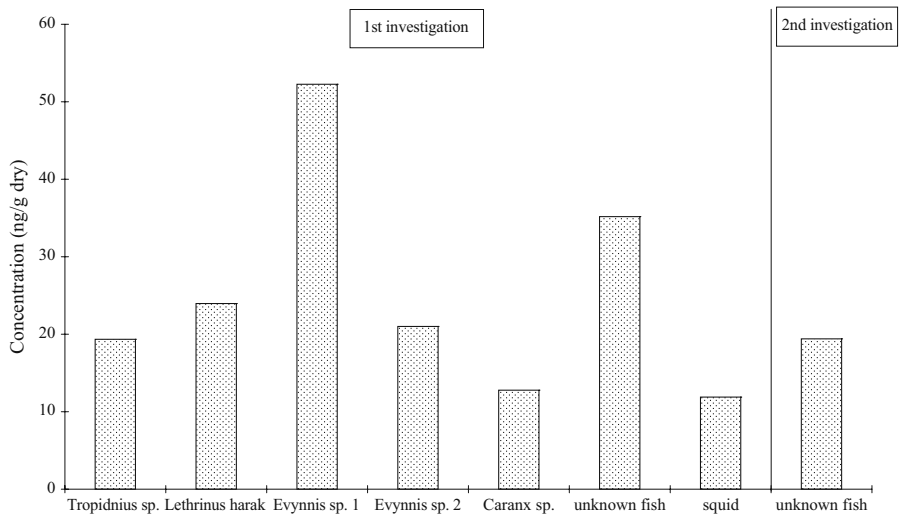


Fig. 2 Concentrations of total PAHs in fish at Luzaran on August (first investigation) and September (second investigation) 2006

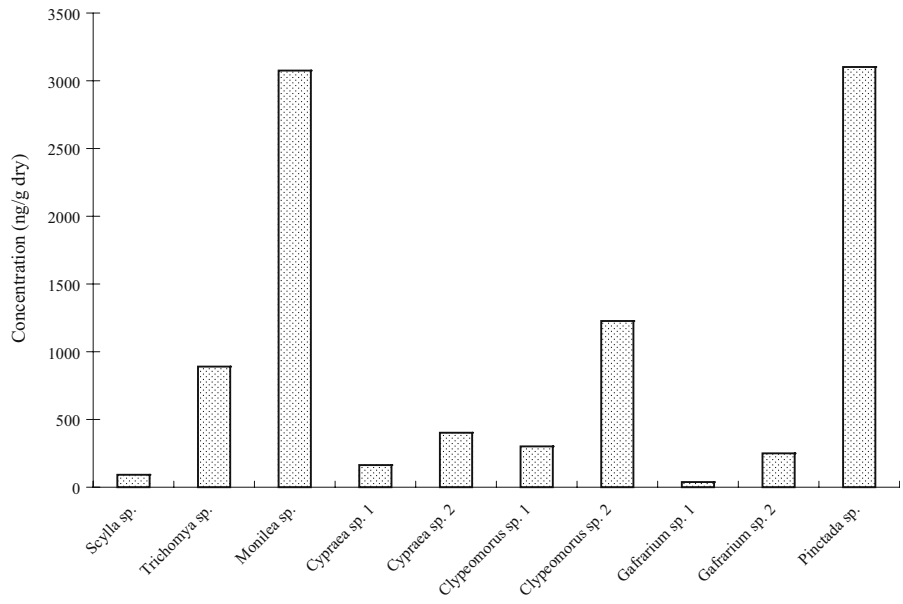


Fig. 3 Concentrations of total PAHs in shellfish and crab at Luzaran on September 2006. The concentrations of *Scylla* sp., *Trichomya* sp., *Monilea* sp., and *Pinctada* sp. were from a single analysis only. However, the sample of *Scylla* sp. was from three individuals and then analyzed. Similarly, *Trichomya* sp. was from 23 individuals, *Monilea* sp. from 18, and *Pinctada* sp. from eight

at sites most affected by the *Prestige* oil spill after 2–3 months. In the *Prestige* oil spill, several effects on shellfish and other aquatic organisms were reported (Cajaraville et al. 2006; Laffon et al. 2006; Pérez-Cadahía et al. 2004). Although those concentrations were slightly higher than those in the *Solar I* oil spill, it could be anticipated that shellfish could be affected by heavy pollution of the spilled oil from *Solar I*. In the *Nakhodka* oil spill, total PAHs (eight PAHs) in spiny top shell (*Turbo cornutus*) were 44.0 ng/g wet weight (Koyama et al. 2004), and the *Solar I* oil spill could have a more serious effect on shellfish than the *Nakhodka*.

In the individual PAH accumulations, pyrene, phenanthrene, and fluoranthene were relatively accumulated in higher concentrations in most fish, although those concentrations were less than 10 ng/g dry weight (the representative example shown in Fig. 5). The low molecular PAHs of

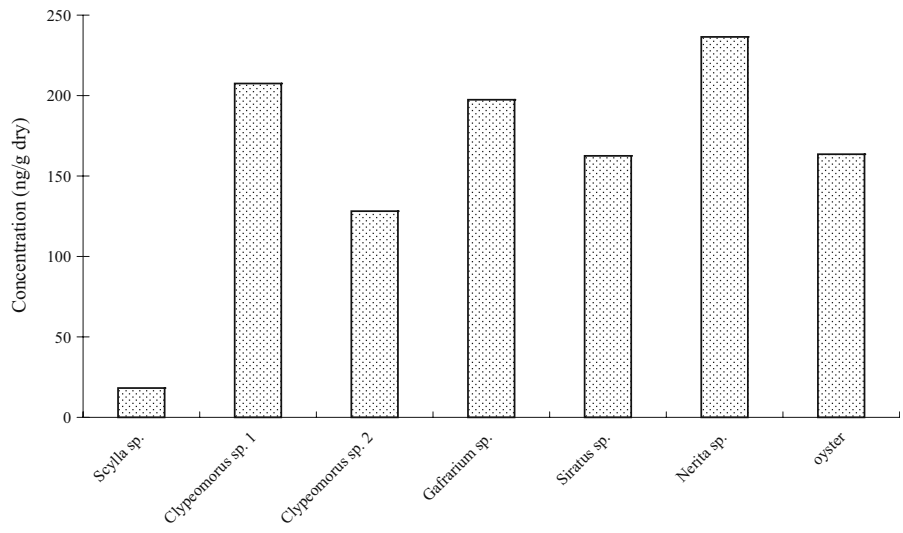


Fig. 4 Concentrations of total PAHs in shellfish and crab at Taklong Island on September 2006. The concentrations of *Scylla* sp., *Gafrarium* sp., *Siratus* sp., *Nerita* sp., and oyster were from a single analysis only. However, the sample of *Scylla* sp. was from two individuals and then analyzed. Similarly, *Gafrarium* sp. was from nine individuals, *Siratus* sp. from two, *Nerita* sp. from 31, and oyster from three

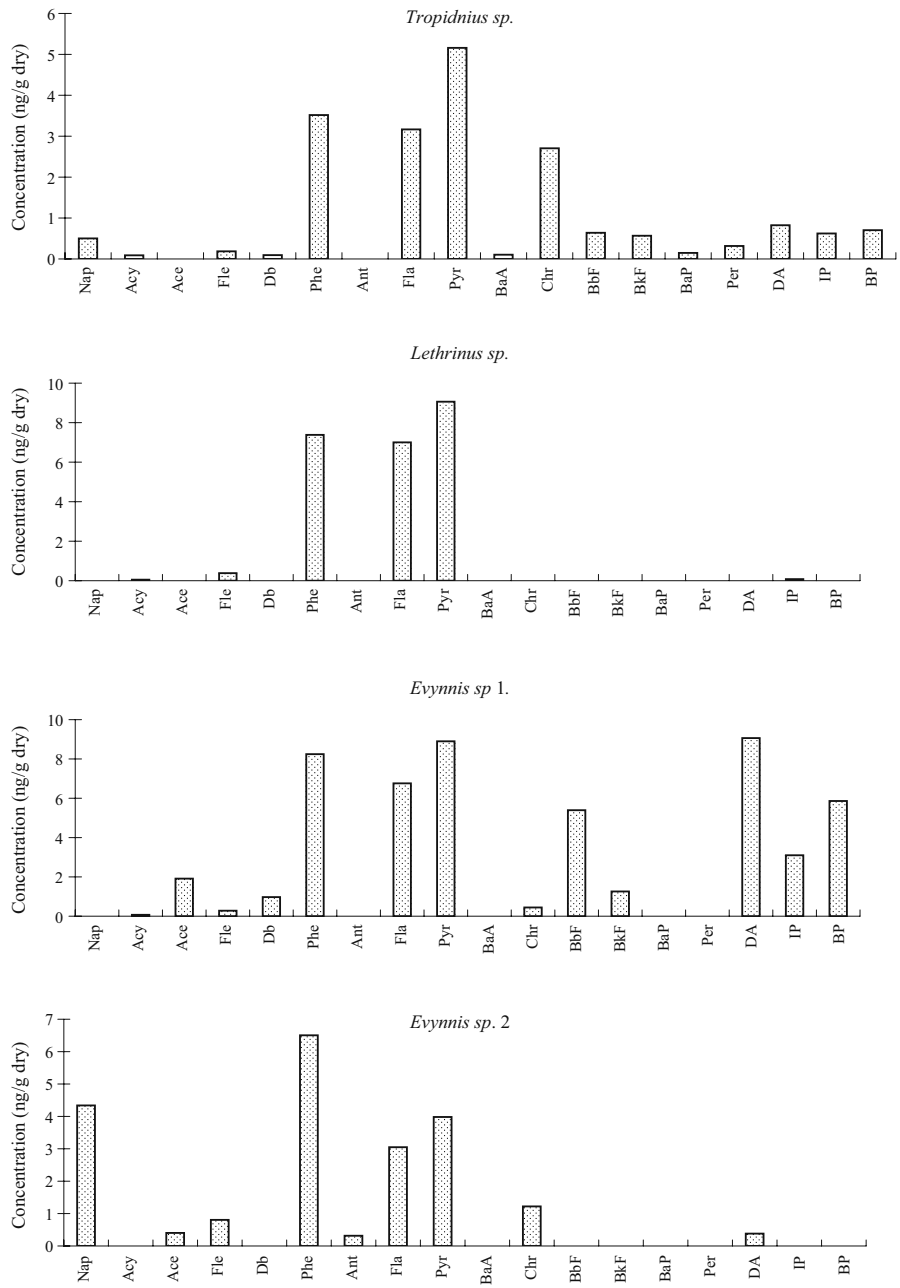


Fig. 5 Concentrations of individual 1 PAHs in fish at Luzaran on August 2006. *Nap* naphthalene, *Acy* acenaphthylene, *Ace* acenaphthene, *Fle* fluorine, *Db* dibenzothiophene, *Phe* phenanthrene, *Ant* anthracene, *Fla* fluoranthene, *Pyr* pyrene, *BaA* benzo(*a*)anthracene, *Chr* chrysene, *BbF* benzo(*b*)fluoranthene, *BkF* benzo(*k*)fluoranthene, *BaP* benzo(*a*)pyrene, *Per* perylene, *DA* dibenz(*a*,*h*)anthracene, *IP* indeno(1,2,3-*cd*)pyrene, *BP* benzo(*g*,*h*,*i*)perylene

naphthalene, acenaphthene, acenaphthylene, fluorene, and dibenzothiophene were hardly detected in fish. Those could have been rapidly metabolized or their concentrations in the spilled oil could be very low. The concentrations of high molecular weight PAHs were different among each fish. These differences could be caused by the following assumptions: (1) Some fish species could escape to lower pollution sites, and other stayed on or near the pollution site. (2) As reported by Cheikyula et al. (2008), there is inter-species difference in the metabolizing ability between fish species. The first assumption corroborated the differences of total PAH concentrations and PAH distribution in the body between *Evynnis* sp. 1 and 2 (Fig. 5).

Chrysene was accumulated in significantly high concentrations in all shellfish at Luzaran and Taklong (Figs. 6 and 7). It was also found during the *Aegean Sea*, *Prestige*, and *Nakhodka* oil spills (Koyama et al. 2004; Porte et al. 2000; Soriano et al. 2006). The individual concentrations of the high molecular weight PAHs for fluoranthene, pyrene, benzo(*a*)anthracene, benzo(*b*)fluoranthene, benzo(*k*)fluoranthene, benzo(*a*)pyrene, perylene, dibenz(*a*,*h*)anthracene, indeno(1,2,3-*cd*)pyrene, and benzo(*g*,*h*,*i*)-perylene also tended to be roughly similar in shellfish, although low molecular weight PAHs were hardly detected. In this investigation, we

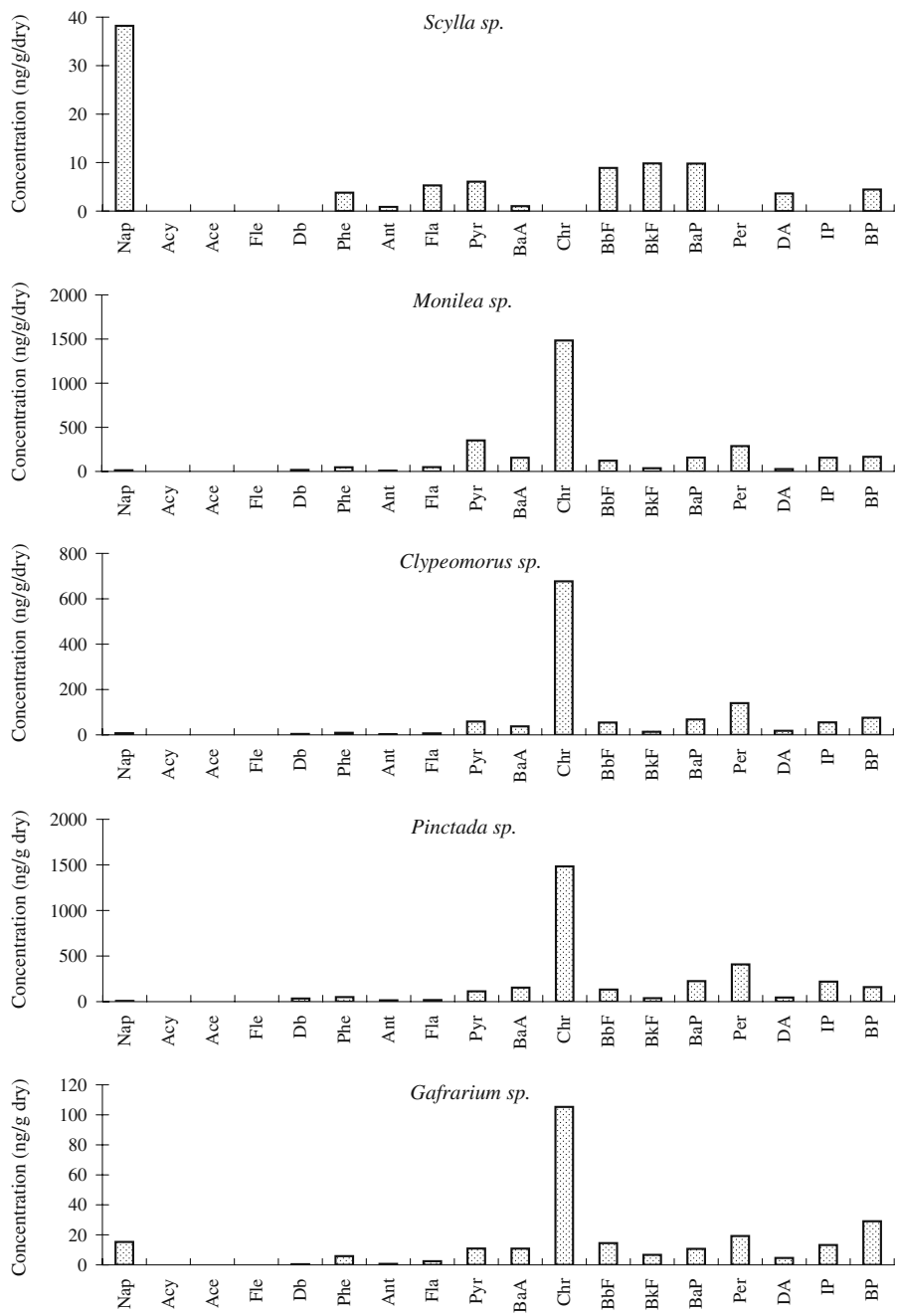


Fig. 6 Concentrations of individual PAHs in shellfish and crab at Luzaran on September 2006. The abbreviations were the same as shown in Fig. 5

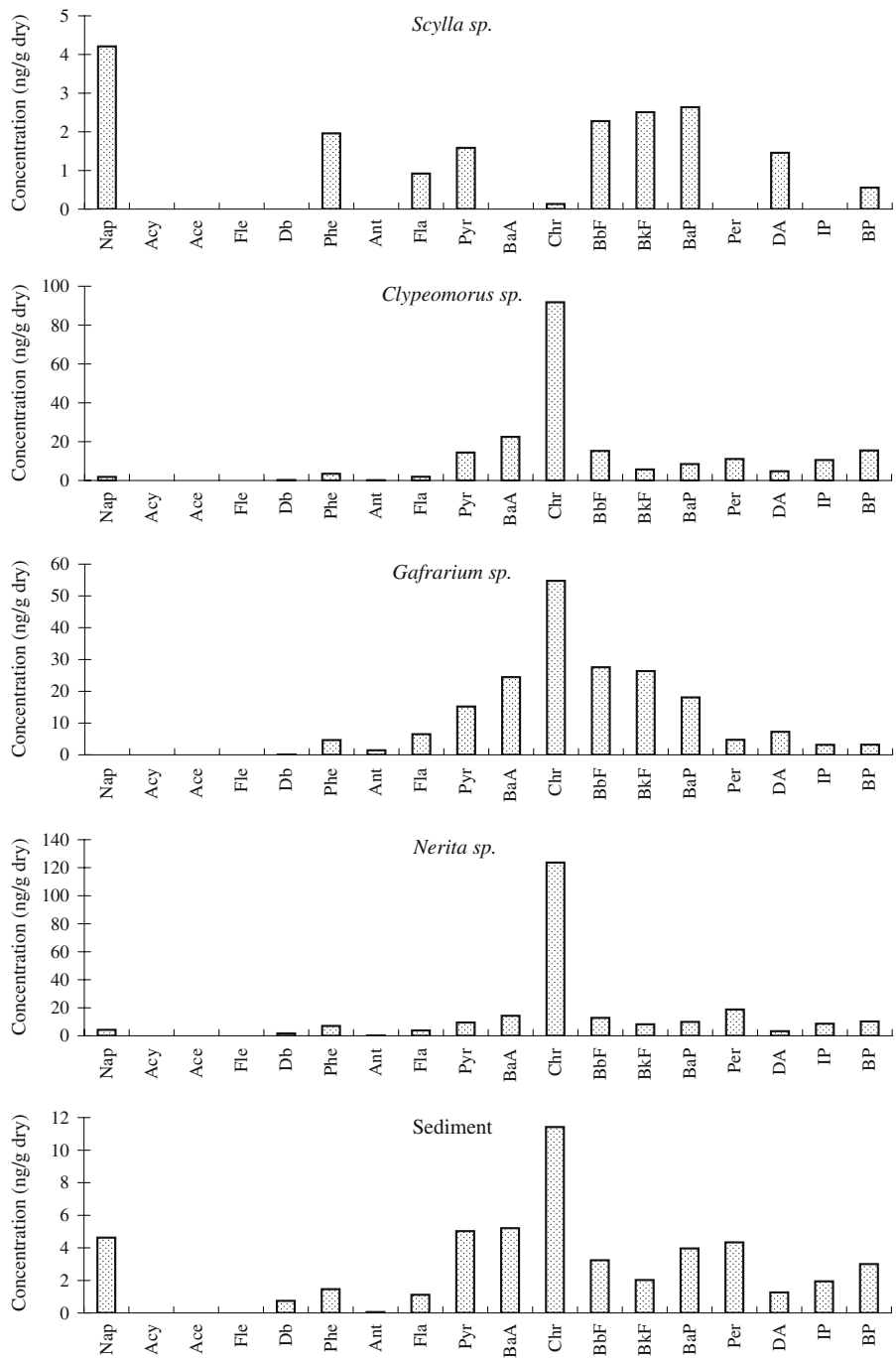


Fig. 7 Concentrations of individual PAHs in shellfish, crab, and sediment at Taklong Island on September 2006. The abbreviations were the same as shown in Fig. 5

could collect only one sediment sample from Taklong, and the PAH distribution pattern roughly corresponded with that in shellfish. The facts provide evidence that shellfish is an excellent bio-indicator organism for oil spill.

The accumulation pattern for individual PAHs was roughly similar in the crab, *Scylla* sp., from both sites, although the concentrations were different. Naphthalene, phenanthrene, fluoranthene, pyrene, benzo(*b*)fluoranthene, benzo(*k*)fluoran-

thene, benzo(*a*)pyrene, dibenz(*a*,*h*)anthracene, and benzo(*g*,*h*,*i*)perylene were commonly detected in crabs at Luzaran and Taklong. The concentrations of naphthalene were highest among PAHs in crabs, although these were found to be much lower in fish and shellfish. On the other hand, chrysene was detected in crabs in Taklong only. Benzo(*b*)fluoranthene, benzo(*k*)fluoranthene, and benzo(*a*)pyrene were also detected in crabs, and their concentrations were relatively high.

Alkylated PAHs

Sometimes fingerprintings for oil pollution were indicated with the petroleum-specific alkylated (C1–C4) homologs of selected PAHs (that is, alkylated naphthalene, fluorene, dibenzothiophene, phenanthrene/anthracene, pyrene/fluoranthene, and chrysene/benzo(*a*)anthracene series). These alkylated PAH homologs are the backbone of chemical characterization and identification for oil spill assessments (Wang et al. 1999). In this investigation, significantly high concentrations of these alkylated homologs were detected especially in shellfish. Recently, risks to human health from exposure to alkylated PAHs in seafoods were evaluated (Baird et al. 2007). Since many people live near the sea in Guimaras Island and depend on seafoods, it will be important to monitor the variations of alkylated PAHs in these aquatic organisms, especially in shellfish.

Alkylated naphthalenes tended to be in higher concentrations in fish than other alkylated PAHs (Fig. 8). C4-dibenzothiophene and alkylated chrysene/benzo(*a*)anthracene were not detected in any fish. C2- and C3-pyrene/fluoranthene were also detected only in four and two fish species, respectively. These seemed to be caused by inhibition of accumulation by the steric effects of alkylated pyrene/fluoranthene and chrysene/benzo(*a*)anthracene (Barron 1990). The accumulations for naphthalenes, fluorenes, and phenanthrenes/anthracenes were elevated such as C0<C1<C2; however, those of C3 and C4 were gradually decreasing from C2 in fish. Also, C1-dibenzotiophene was of higher concentration than

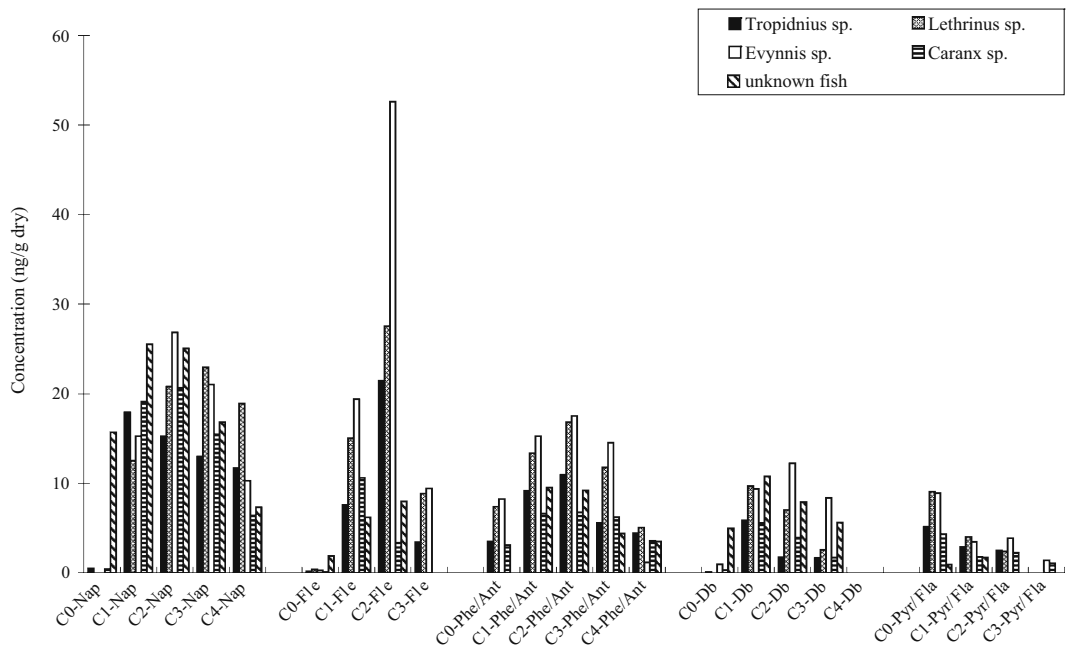
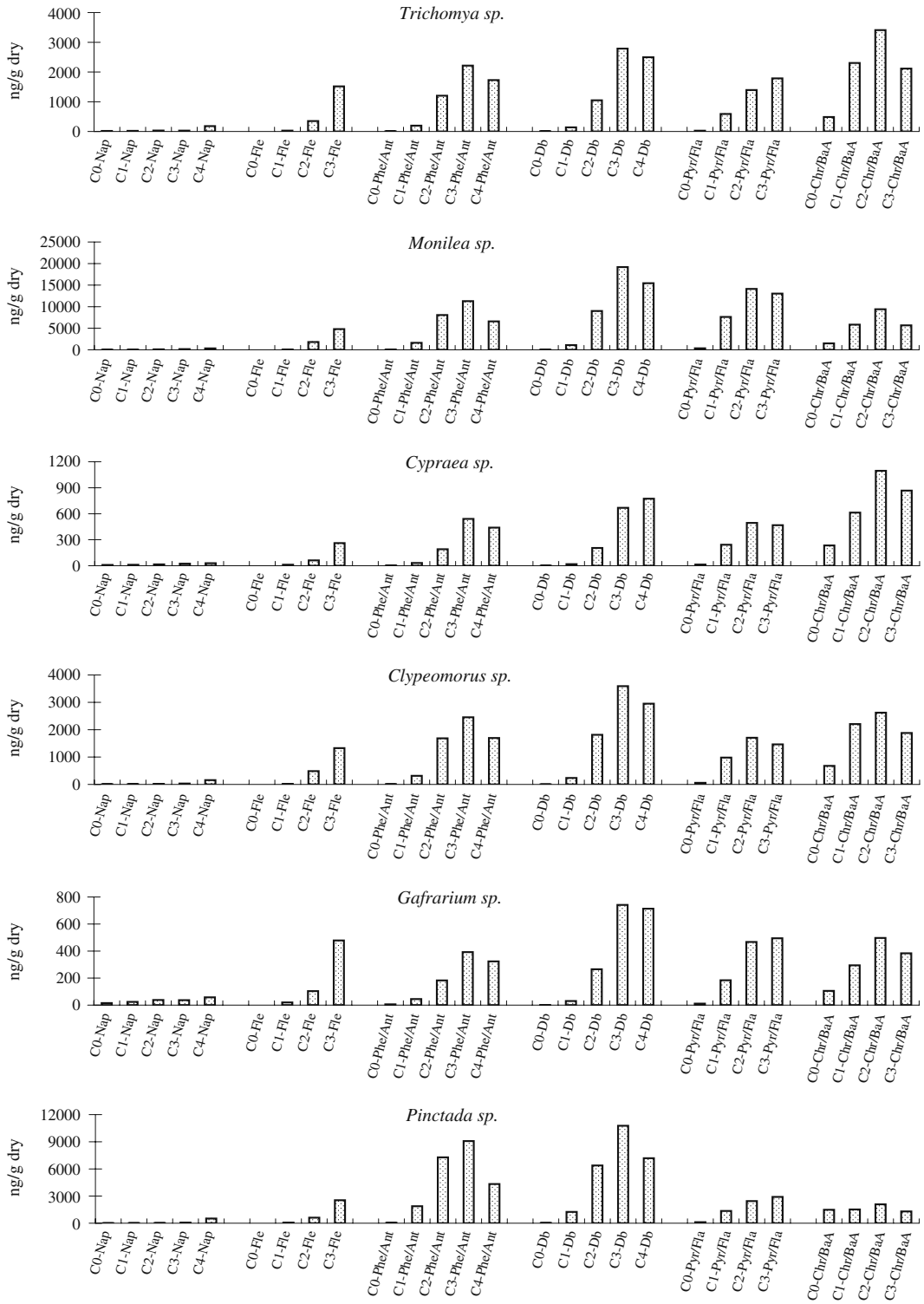


Fig. 8 Concentrations of individual alkylated PAHs in fish at Luzaran on August 2006. The abbreviations were the same as shown in Fig. 5


Trichomya sp.
Monilea sp.
Cypraea sp.
Clypeomorus sp.
Gafrarium sp.
Pinctada sp.
ng/g dry
C0-Nap
C1-Nap
C2-Nap
C3-Nap
C4-Nap
C0-Fle
C1-Fle
C2-Fle
C3-Fle
C0-Phe/Ant
C1-Phe/Ant
C2-Phe/Ant
C3-Phe/Ant
C4-Phe/Ant
C0-Db
C1-Db
C2-Db
C3-Db
C4-Db
C0-Pyr/Fla
C1-Pyr/Fla
C2-Pyr/Fla
C3-Pyr/Fla
C0-Chr/BaA
C1-Chr/BaA
C2-Chr/BaA
C3-Chr/BaA

◀ **Fig. 9** Concentrations of individual alkylated PAHs in shellfish at Luzaran on September 2006. The abbreviations were the same as shown in Fig. 5

the parent dibenzothiophene, and the accumulation was C1>C2>C3. Parent pyrene was higher than all alkylated homologs. The most alkylated PAH concentrations were less than 30 ng/g dry weight.

The accumulation for alkylated naphthalenes and fluorenes in shellfish was of the order of C0<C1<C2<C3(<C4), respectively. Phenanthrenes/anthracenes and dibenzothiophenes were also elevated from C0 to C3, and C4-homologs were lower in concentration than C3. For pyrenes/fluoranthenes and chrysenes, their accumulations varied such as C0<C1<C2, and C3 was similar with or decreasing from C2. At Luzaran, the concentrations of alkylated PAHs were relatively high, and especially, alkylated phenanthrenes/anthracenes, dibenzothiophenes, and pyrenes/fluoranthenes in *Monilea* sp. and *Pinctada* sp. were remarkably of high concentrations (Fig. 9). At Taklong, their concentrations in shellfish were lower and less than 1,000 ng/g dry weight (Fig. 10). The distribution patterns were similar between sediment and shellfish; however, the concentrations in sediment were relatively lower than those in shellfish (Fig. 10). This suggested that PAHs and alkylated PAHs in shellfish were derived from PAHs and alkylated PAHs adsorbed onto suspended substances which are finally deposited in sediments. Certain diagnostic ratios have been widely used in identifying sources of PAHs, and petrogenic sources are characterized by high ratios of mono-methylphenanthlene (MP)/phenanthrene (P) (MP/P > 2; Fang et al. 2003). In Taklong, MP (1-methylphenanthlene)/P in sediment and shellfish were as follows: 4.8 in sediment, 2.1 in *Clypeomorus* sp., 6.1 in *Gafrarium* sp., 3.5 in *Siratus* sp., 1.4 in *Nerita* sp., and 2.6 in oyster. As a result, PAHs and alkylated PAHs in sediment and shellfish at Taklong, except for *Nerita* sp., could be derived from the spilled oil. Since there has not been heavy traffic and high industrial activity in Guimaras Island, we are convinced that alkylated PAHs in shellfish were accumulated from the *Solar I* oil spill at Luzaran and Taklong, although their concentrations were different between the two sites. In comparison with fish, shellfish could continue to be polluted by heavy oil.

Among alkylated PAHs with a few exceptions, C3-dibenzothiophene was roughly highest in shellfish. This tendency in shellfish was more notable than in fish and crab. The concentration in shellfish at Taklong was from 29.0 to 708 ng/g dry weight and that at Luzaran was from 58.6 to 19,186 ng/g dry weight. Luzaran was, therefore, more heavily polluted by the *Solar I* oil spill than Taklong. It has been pointed out that sulfur heterocyclic is less degraded than PAH derivatives (Meyer and Steinhart 2000). Alkylated dibenzothiophenes could continue to pollute around Guimaras Island for some time in the future, as what Koyama and Kuroshima (1998) had observed of the persistency of alkylated dibenzothiophenes 3 years after *Gulf War* oil spill.

The distribution patterns and concentrations of PAHs and alkylated PAHs in shellfish were completely different from those of fish. The difference had two possible causes. Firstly, the difference could be attributed to the uptake route of PAHs and alkylated PAHs between fish and shellfish. The main route of fish could be from water-soluble heavy oil. On the other hand, that of shellfish could be from suspended substances. In general, water solubility of PAHs is getting lower with increasing benzene ring, and alkylated pyrene/fluoranthene and chrysene/benzo(*a*)anthracene expected much low solubilities in water. Therefore, alkylated pyrene/fluoranthene and chrysene in fish were significantly lower than other alkylated PAHs (Fig. 8). Secondly, shellfish have lower rates of PAH metabolism than fish (Stegeman and Lech 1991). As shown in Fig. 10, in sediments, C3-homologs had the highest concentration in alkylated PAH, and spilled oil could also contain most C3-homologs. As pointed out above, the distribution of alkylated PAHs was similar between sediment and shellfish. Furthermore, the concentrations in shellfish were higher than those in sediment. These facts suggested that alkylated PAHs were slowly eliminated from shellfish. On the other hand, C2-homologs of naphthalene, fluorene, phenanthrene, and dibenzothiophene

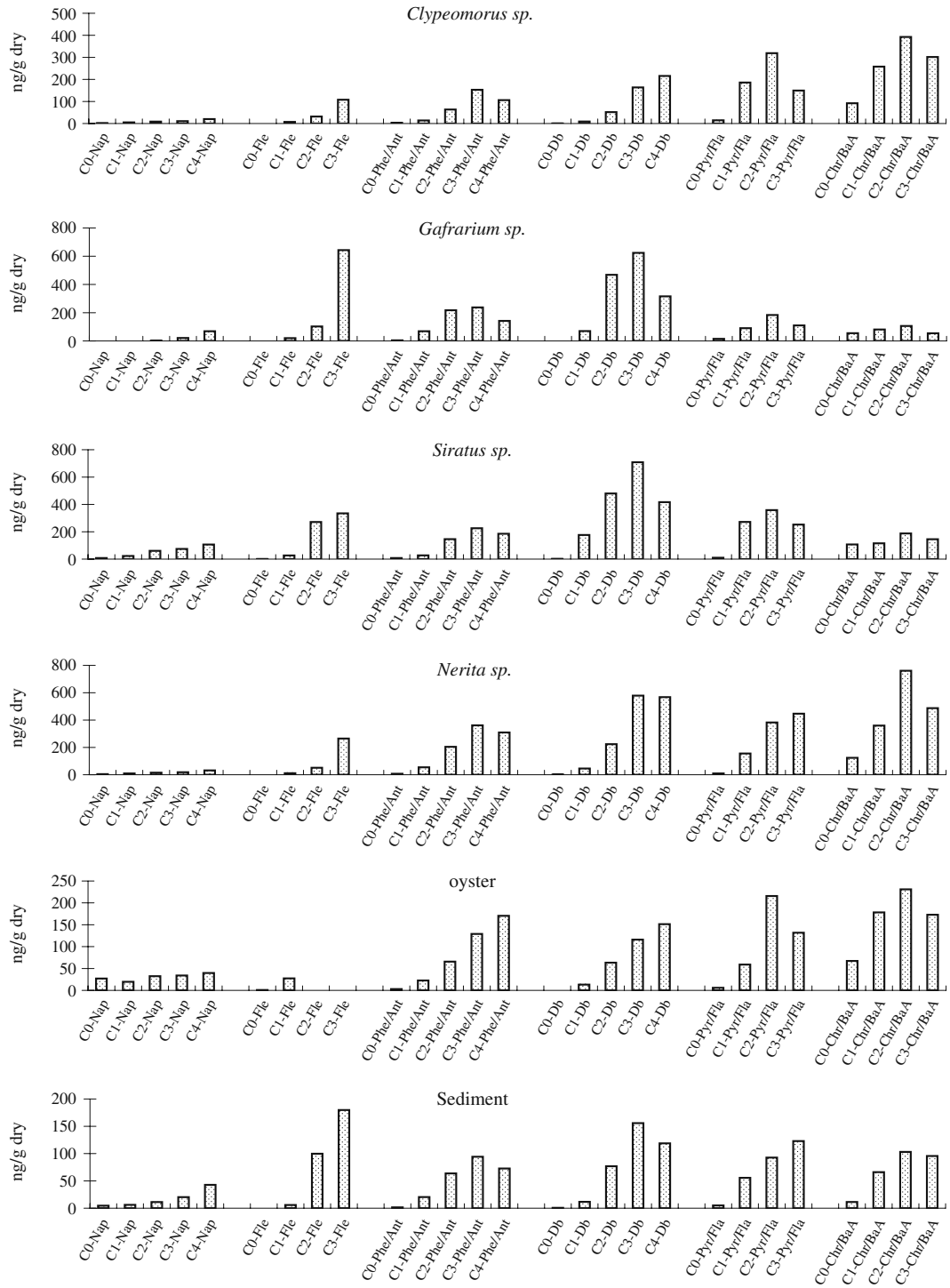


Fig. 10 Concentrations of individual alkylated PAHs in shellfish and sediment at Taklong Island on September 2006. The abbreviations were the same as shown in Fig. 5

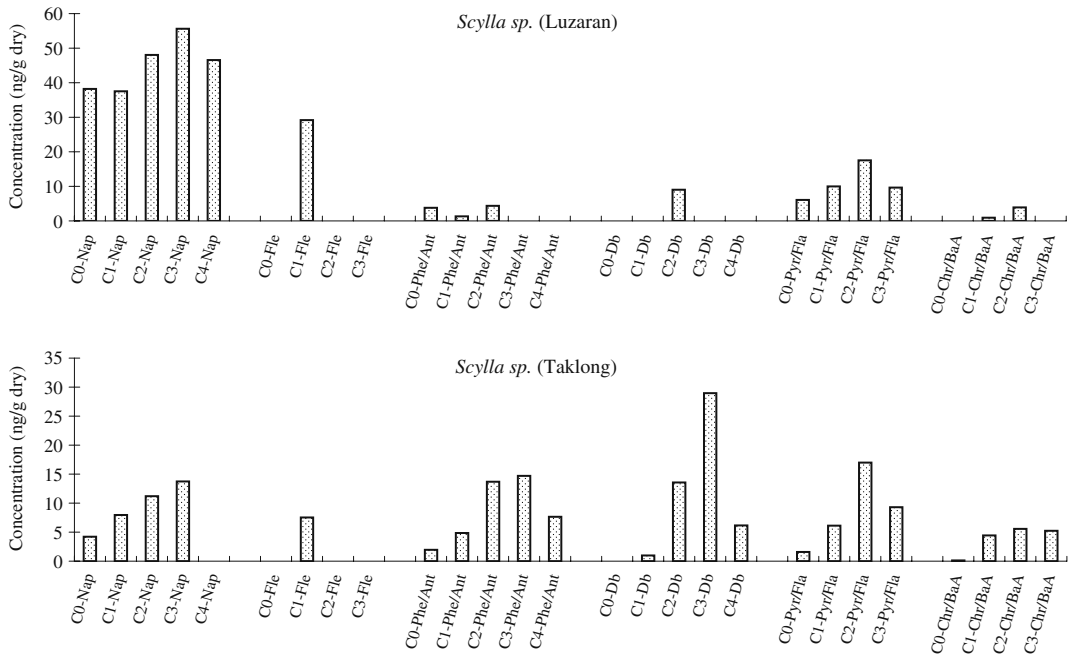


Fig. 11 Concentrations of individual alkylated PAHs in crab at Luzaran and Taklong Island on September 2006. The abbreviations were the same as shown in Fig. 5

and the parent pyrene/fluoranthene (C0) had the highest concentrations in fish, respectively, and the concentrations of most alkylated PAHs were lower than those in shellfish. While the metabolite profiles in shellfish are mainly quinones (Michel et al. 1995; Stegeman 1985), those of fish are hydroxylated derivatives (Deshpande 1989; Krahn et al. 1987; Krone et al. 1992; Mazeas and Budzinski 2005; Stein et al. 1990; Uno et al. 2003). The difference of the metabolic system between fish and shellfish could be caused by the difference in their metabolizing abilities.

The distribution patterns of alkylated PAHs in crabs were different between Luzaran and Taklong (Fig. 11). Crabs move widely and so their accumulated patterns could not be reflected in the collected sites.

Conclusion

The *Solar I* oil spill was one of the worst catastrophes at sea in the Philippines. This study, which reported high concentrations of PAHs and alkylated PAHs in shellfish, is only an initial report on the pollution caused by the *Solar I* oil spill. There were "hot spots" with heavy oil pollution, and many people who collected and consumed the aquatic organisms in the coastal area of Guimaras may have been exposed to health risks. The investigation of this pollution needs to be continued to provide further information on the extent of contaminations.

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